Michael Crandall

856-685-8737 | mcrandal@terpmail.umd.edu | linkedin.com/in/crandall0327 | github.com/hiitsmike1
Cinnaminson, NJ

EDUCATION

University of Maryland, College Park – College Park, MD December 2028
Bachelor of Science, Electrical Engineering
Cumulative GPA: 4.0

TECHNICAL SKILLS

Hardware & Embedded: Discrete circuit design, KiCad, STM32 and ESP32 microcontrollers, Arduino, Zephyr RTOS, CAN
Languages: C, C++, Python, Verilog, Java, MATLAB, HTML
Tools: Onshape, GTKWave, CAD, Git/GitHub, Microsoft Word & PowerPoint, Google Drive

RELEVANT PROJECTS

4-bit ALU Design Using RTL – Personal Project 2026
- Designed, built, and tested a 4-bit ALU entirely from discrete components using resistor-transistor logic.
- Used 160+ individual NPN BJT transistors and 270 resistors to implement 7 distinct hardware operations, including Addition, Subtraction, bitwise OR/AND/NOT, and Shift Left/Right (SHL/SHR).
- Designed a multiplexer and tri-state output stage to safely route from all 7 operations onto a shared 4-bit bus.

8-bit CPU Design in Verilog - Personal Project 2026
- Designed a custom instruction set architecture supporting LOAD, STORE, and ADD operations across a 4-register file and 32-word RAM.
- Implemented a multi-cycle finite-state-machine control unit to sequence instruction fetch, decode, and execute phases over a shared tri-state-emulated data bus.
- Debugged bus contention and timing hazards caused by overlapping datapath signals.

Autonomous Mission Robot – Team Project – College Park, MD 2025
- Collaborated in an 8-person team to design and build a robot that autonomously navigated an obstacle course, planted a seed, and collected rock samples.
- Implemented sensor-based obstacle detection and control logic using an Arduino microcontroller.
- Assisted with hardware wiring, motor driver integration, and testing to complete the mission.
- Applied a machine learning model to identify different orientations with 80% accuracy, improving reliability.

Multitasking Firmware on ESP32 with Zephyr RTOS – Personal Project 2026
- Built firmware that runs four concurrent tasks on an ESP32 (buzzer, servo, LED, and serial data reception from an Arduino), scheduled under Zephyr RTOS.
- Wrote custom out-of-tree driver modules for servo (PWM) and GPIO LED control, configured through device tree.

ALU Design in Verilog – Personal Project 2025
- Designed a Verilog ALU verified with a custom testbench of 327,680 cases and a GTKWave simulation

EXPERIENCE

University of Maryland College Park, MD
Undergraduate Teaching Fellow (UTF) – ENEE140 August 2026 – Present
- Facilitate twice-weekly discussion sections for 14 undergraduate students, reinforcing coding concepts.
- Hold twice-weekly office hours to support students and review material.
- Grade homework assignments, projects, and exams, providing accurate and constructive feedback.

Lavner Education Cherry Hill, NJ
STEM Instructor/Intern June 2026 – August 2026
- Taught students aged 6-14 advanced STEM concepts in robotics, coding, physics, and game development using proprietary curricula.
- Helped manage inventory and material allocation across 10 classes to guarantee resource availability for curricula.

Boyars Food Market Ocean City, NJ
Deli Associate Summers 2023 & 2024
- Prepared and packaged customer orders to high standards of cleanliness and accuracy; stayed organized and communicated clearly during busy rushes.

LEADERSHIP & INVOLVEMENT

UMD Loop - IEEE-HKN - Engineering Honors Program - DoQuantum -
- Planning rover wireless communication for URC4: radio selection, directional base station antenna, and test plan.
- Revised an STM32F103 schematic in KiCad: 8 MHz crystal/load caps (72 MHz via PLL), LT1084 3.3 V regulator.

Cinnaminson High School — Robotics Club Member (2021–2025), NHS Officer (2024–2025)
St. Charles Borromeo — Volunteer, event support and operations (Cinnaminson, NJ)